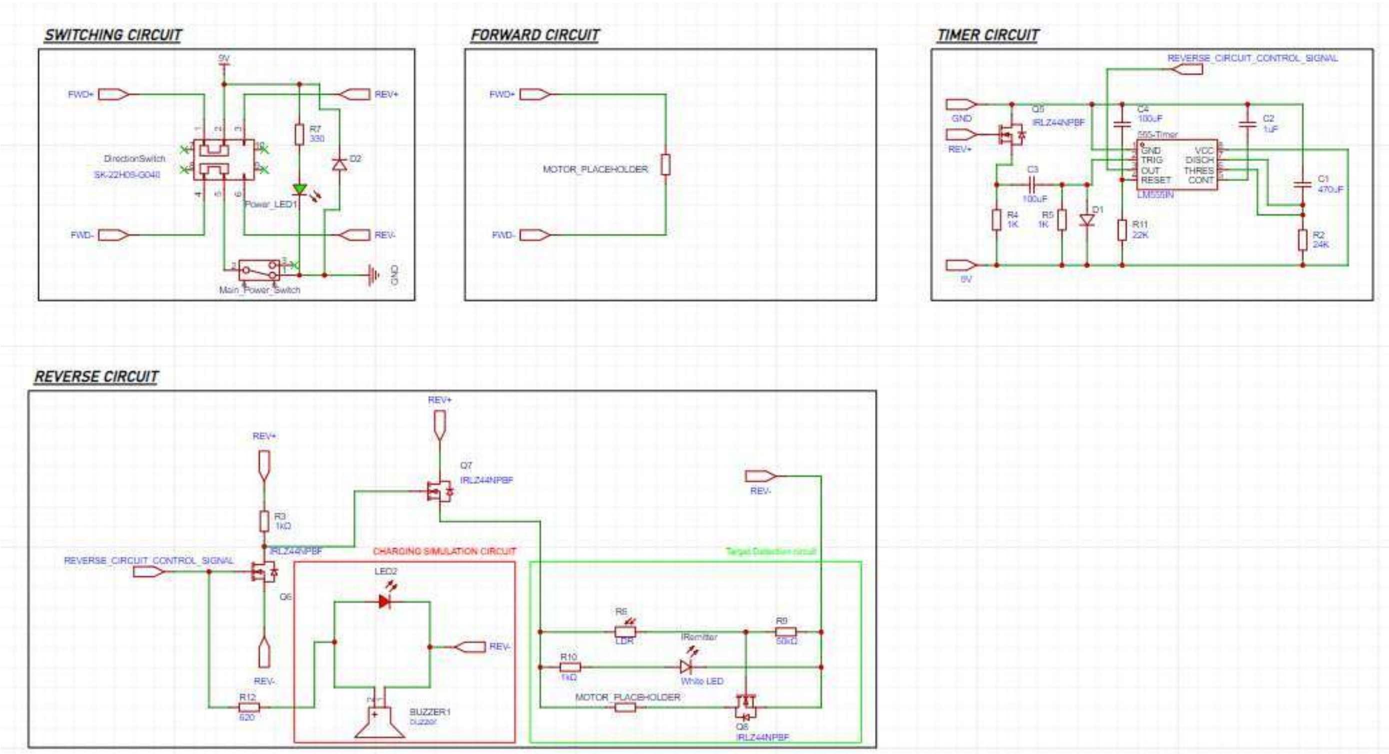
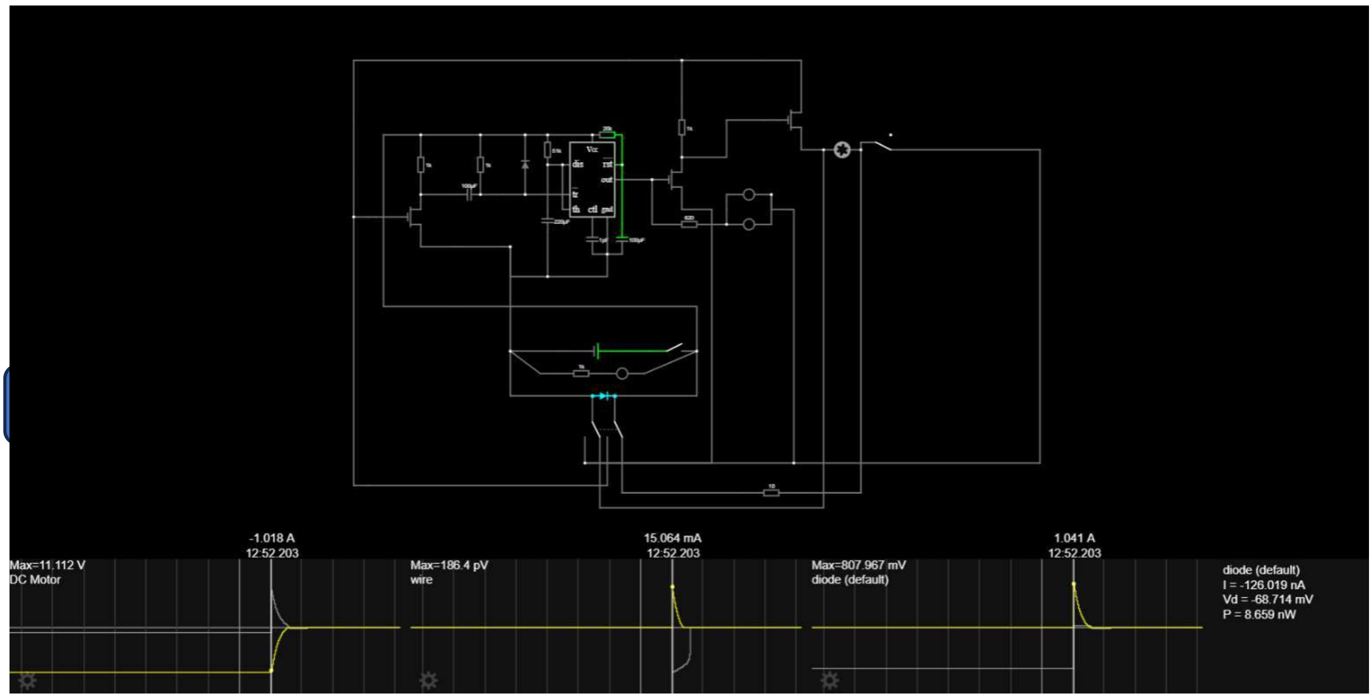
- Based on the circuit testing above, an updated schematic was developed as seen below



- **Alteration 1:** Added an RC circuit in parallel with the RESET pin of the 555 timer to prevent false triggers on startup by holding RESET low for a short period.
- **Alteration 2:** Added a diode in parallel with the main power rail to counteract flyback voltage spikes generated from the motor.
- **Alteration 3:** Removed MOSFET Q4 (see V1 schematic) as it was redundant.

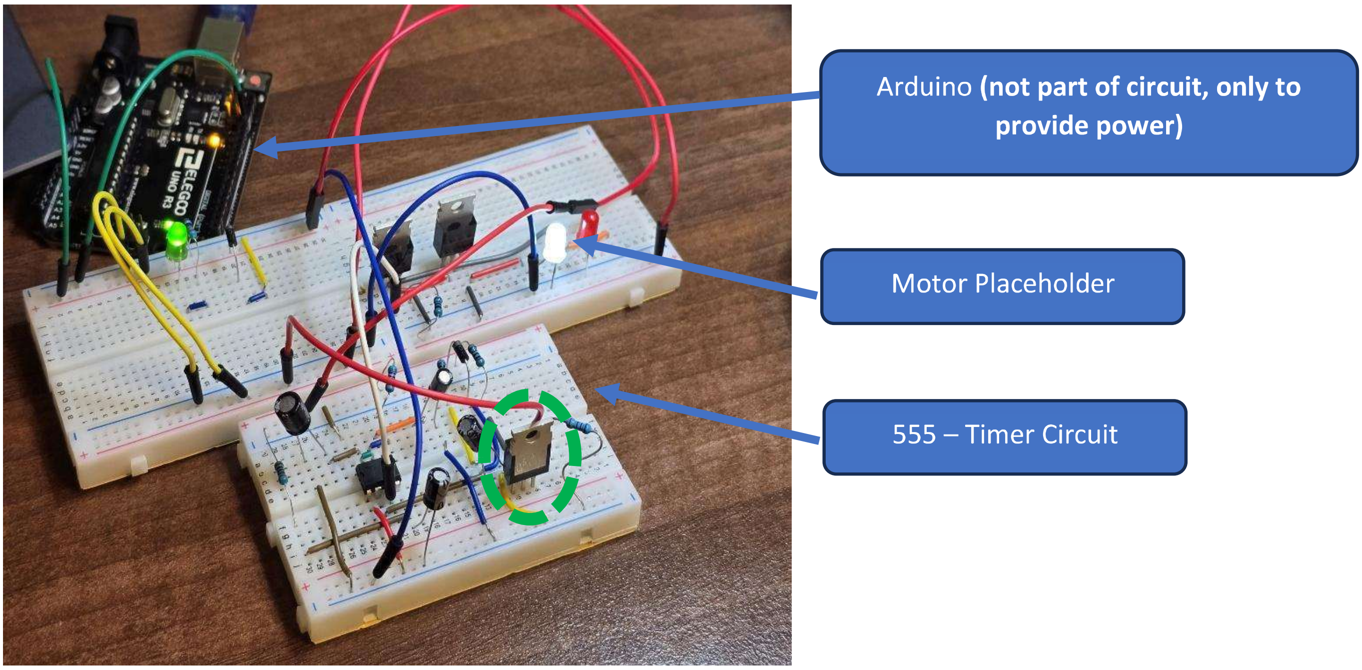
Simulating Circuit V2:

- Below is the Falstad simulation of the updated circuit (wiring paths adjusted for clarity)



V2 Circuit Simulation Results:

- LED and timing sequences activated as per the specification
- 555-timer no longer falsely triggers upon startup
- Large voltage spike now reduced to <1V with a max current draw of 1.041A during flyback.



Testing a prototype of Circuit V2 (see above for image):

- To test the circuit's real-world performance, we constructed a rough prototype using breadboards and available components.
- Due to the availability of certain components, placeholders were used. For example, the motor was replaced with a white LED. Additionally, the DPDT switch was replaced with a manual switching system which involved changing the power rail from one circuit to another.
- Some component values had to be altered to suit what was available.
- Finally, it is important to note that a 5V power system (provided by the Arduino) was used instead of the planned 12V. This was due to the lack of 12V power available to us during testing. However, the circuit principles should remain largely compatible with both voltages.

V2 Prototype Circuit Testing Results:

- Timer behaved as expected and was reliable.
- The triggering of the timer exhibited issues. Primarily, it was noted that to trigger the timer a second time, the trigger pin would need to be grounded before connected to the power rail. This was unexpected behaviour, and was researched to ensure no errors

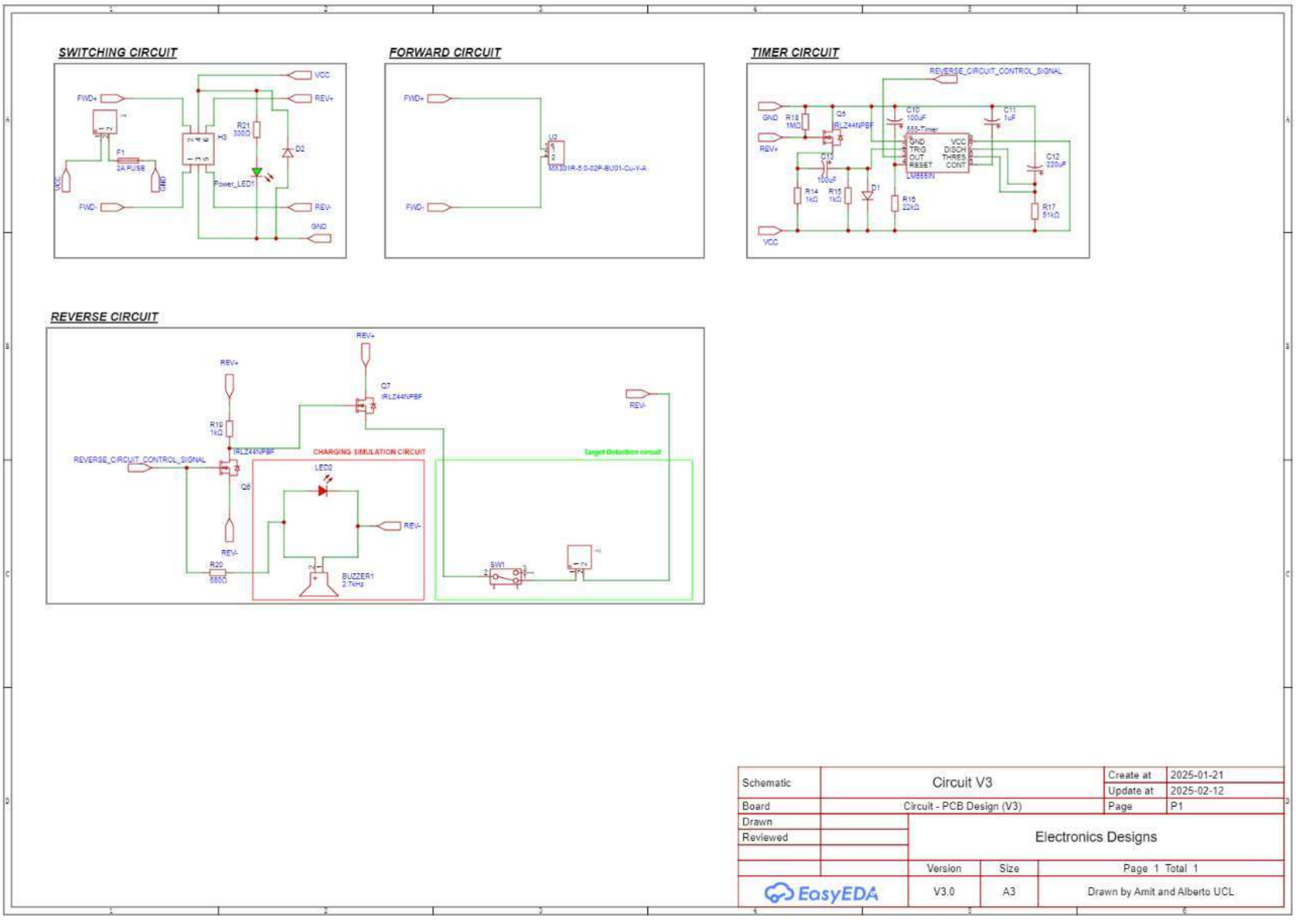
Identifying and Resolving the Root Cause of the Issue:

- In order to identify the cause of the issue above, the circuit was probed extensively with a multi-meter. The cause was then identified to be the MOSFET Q5 (Highlighted in green on the breadboard). The gate of the MOSFET fails to return to LOW even after the power supply is removed.

- In order to rectify this, we added a 1 Mega-Ohm pull-down resistor. This ensured the gate was tied to ground at all points.
- Further testing with this solution showed that the circuit now behaved fully as intended and simulated.

Updating the Schematic to Reflect on Testing Changes (V2 -> V3):

- The schematic below is designed to reflect on how the circuit would be built in the real-world as opposed to in theory.

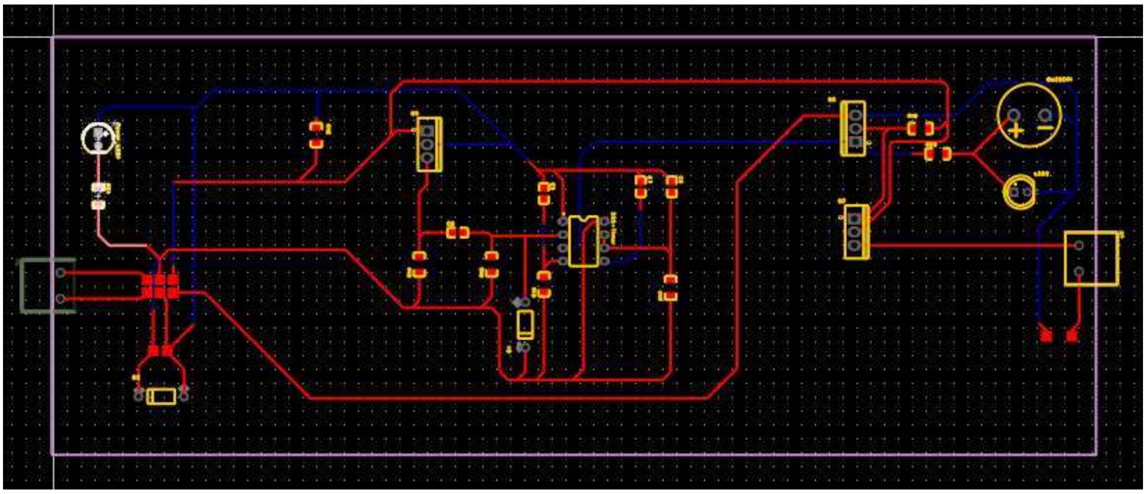


- **Alteration 1:** Adjusted all capacitor values to match commercially available components.
- **Alteration 2:** Replaced all capacitors with polarized capacitors to match what will be used in the real circuit
- **Alteration 3:** Added the 1 Mega-Ohm pull down resistor between the gate of MOSFET Q5 and GND
- **Alteration 4:** Replaced switches with pin headers (as switches will be detached from the main circuit and connected via wire extenders)

Developing the Schematic into a PCB (V1):

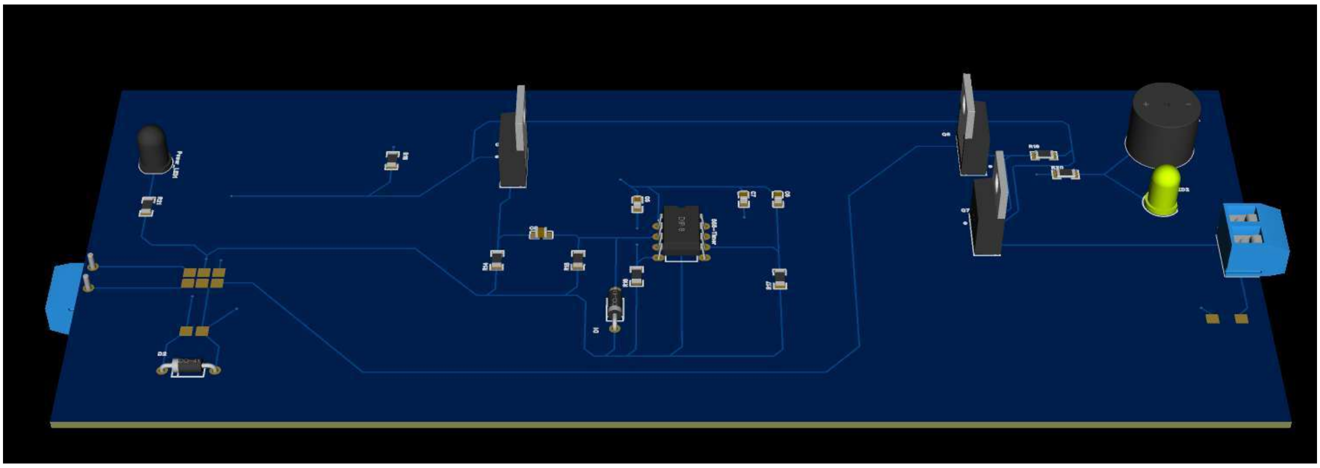
- Whilst breadboards are effective for testing and rapid prototyping, their connections are often unreliable
- A PCB is a far more rigid solution and provides superior reliability.

- Below is an image of the initial PCB layout on EasyEDA



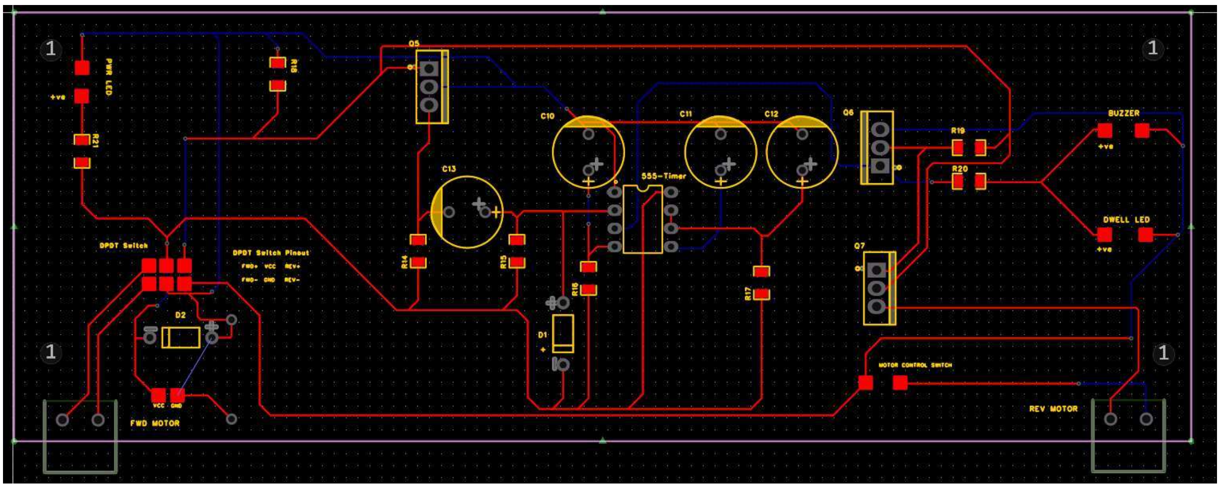
- This layout uses a mix of THT (Through Hole Technology) and SMD (Surface-Mount Device) components.
- Allocated a 200x80mm rectangular space for the components as an initial frame.
- Since the position of the DPDT and SPST switch will be outside the board, copper pads were added to mount wire leads.

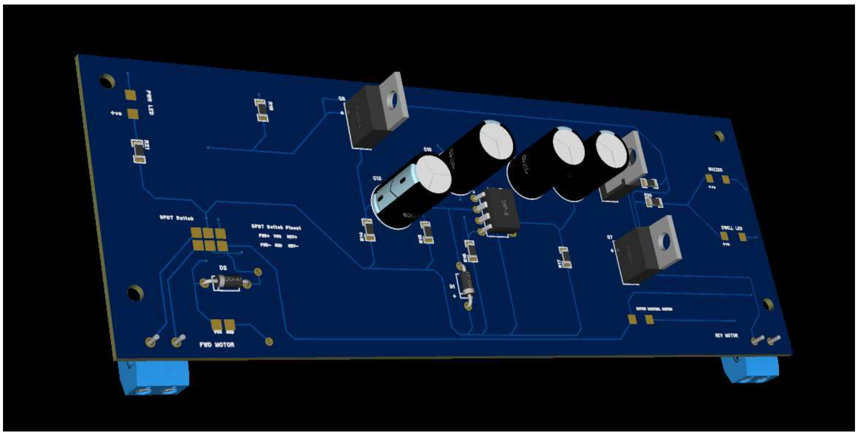
3D Model of PCB (V1):



Developing the PCB (V2):

- The focus of this development is compacting the circuit down. See the updated layout below:





- **Alteration 1:** Swapped all SMD capacitors for THT equivalents (to match available components)
- **Alteration 2:** Added fuse holder in series with the main VCC line
- **Alteration 3:** Compacted the PCB outline from 200x80mm to 165x60mm
- **Alteration 4:** Replaced LED and buzzer positions with copper pads to allow for these components to be placed elsewhere on the frame
- **Alteration 5:** Adjusted position of motor screw terminals to rear of PCB
- **Alteration 6:** Added text to aid with describing the pinouts of all copper pads

Future steps:

- Test circuit with planned motor models to check simulation results are valid
- Test with DPDT switch in position
- Produce, assemble and test PCB.

Strength, Stiffness, Stress